

OrbitCover: Interaction-Balanced Estimation of Semantic Predictor Quotients

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Abstract

Two tables can encode the same supervised problem while differing only in feature order, opaque category identifiers, or target identifiers. A learning pipeline need not return the same predictor on these equivalent representations, and ordinary seed ensembling does not integrate over this larger nuisance space. We formulate the output of a randomized learning pipeline as a vector-valued function on a declared finite product of semantic symmetries and training randomness. Its orbit average is a quotient predictor. For Brier and squared loss, prediction variance around this quotient is exactly the expected finite-member loss overhead. Product functional ANOVA (fANOVA) makes its source observable. OrbitCover applies a randomized mixed-level orthogonal array to the complete train-and-predict pipeline: a strength- t design cancels every fANOVA component of order at most t . We evaluate 12 datasets, three splits, and four neural architectures (144 primary cells). At $B = 16$, the coupled strength-2 construction has lower method-relative residual than canonical independent ensembling in 144/144 cells and 12/12 datasets, with a dataset-balanced reduction of 55.9% and a dataset-clustered 95% interval of [38.7%, 73.8%]. This is not a same-estimand comparison: the canonical and symmetrized expectations are measurably distinct in 10/144 cells. A 48-cell same-target ablation attributes a 47.6% mean cellwise reduction to jointly balancing schema, initialization, and training order. Fresh independent randomness removes the schema-only benefit (−7.0% versus canonical independent), and the strength-2 advantage over simple random sampling vanishes on average at convergence (residual ratio 1.002). OrbitCover is therefore an auditable finite-budget estimator for a declared semantic quotient, not a universally better ensemble.

1 Introduction

Tabular models receive an array, but users intend a statistical problem. Reordering feature blocks, bijectively renaming opaque categories, or swapping class identifiers together with their decoding leaves that problem unchanged. The implementation path is less abstract: positional parameters, random initialization, batching, dropout, and finite optimization can couple to the chosen representation. Two legitimate executions can therefore produce different aligned predictions.

Recent tabular work has made models stronger and evaluation broader. TabM studies parameter-efficient neural ensembling (Gorishniy et al., 2025); TabPFN-v2 performs in-context prediction with a learned prior (Hollmann et al., 2025); TabDPT and TabSTAR develop foundation-style tabular prediction (Ma et al., 2025; Arazi et al., 2025); and TabArena and TabReD emphasize reproducible, realistic evaluation (Erickson et al., 2025; Rubachev et al., 2025). These advances sharpen a basic question: which randomized predictor is a reported score intended to describe? Several seeds of one canonical encoding target only one slice of the pipeline. Averaging arbitrary schema transformations is also incomplete unless outputs are aligned and the sampling law is declared.

We define the target first. A finite nuisance product Ω contains exact schema transformations and stochastic choices of the complete learning pipeline. After semantic alignment, its prediction barycenter is the quotient predictor. Estimating it is finite-dimensional integration of an expensive, vector-valued black box. Randomized orthogonal arrays (OAs) are classical variance-reduction designs (Owen, 1992; Ai et al., 2016). We ask when their low-order cancellation helps when the factors are semantic transformations and stochastic choices of an entire neural training run.

The answer is deliberately qualified. Jointly balancing schema and stochastic factors sharply reduces small-budget quotient-estimation error. Balancing schema while assigning an unrelated fresh RNG does not. At convergence, the relative advantage over sampling without replacement vanishes on average. Symmetrization can also change the target prediction. These failures prevent the false universal story that structured ensembles dominate seed ensembles.

Our contributions are:

1. A complete-pipeline semantic quotient. Exact representation equivalences and training randomness form an explicit product estimand with semantic output alignment and separate canonical, independent-joint, and finite-coupled targets.
2. Exact prediction-space accounting. Brier/MSE overhead equals Hilbert variance around the quotient. Product fANOVA exposes main and interaction components, and a strength- t randomized OA removes every component through order t .
3. A broad falsificatory evaluation. A frozen closure covers 12 datasets, three splits, four neural architectures, independent full-pipeline seeds, realistic training sizes and optimization budgets, matched-function controls, and a same-target coupling ablation.
4. An explicit boundary map. Fresh-RNG schema balance fails, target shift is measurable, convergence erases the mean SRS advantage, and better validation fidelity barely changes held-out selection regret.

We claim novelty for neither OAs, fANOVA, group averaging, nor antithetic sampling. The proposed contribution is their complete-pipeline, semantically aligned composition and an audit that separates target and coupling effects.

2 Semantic nuisance quotients

2.1 Setup

Let $D = (X, y)$ be a training sample and x a fixed evaluation sample. Let $\Omega = \Omega_1 \times \dots \times \Omega_d$ be a finite product of declared nuisance factors. Every $z \in \Omega$ preserves the supervised problem. In our main experiments, schema factors are feature-block order, within-field category-ID maps, and target-ID maps; stochastic factors control initialization and training order. Let

$$P_z(x) = \text{Align}(A(zD; z), zx)$$

be the prediction vector after mapping rows, classes, and regression scale back to canonical coordinates. For a declared law μ on Ω , define

$$Q_\mu(x) = \mathbb{E}_{Z \sim \mu}[P_Z(x)]. \quad (1)$$

This definition is distribution-relative. A canonical seed ensemble, a schema $\times$ fresh-seed ensemble, and a finite coupled schema $\times$ initialization $\times$ order product need not have the same expectation. We denote them Q_{can} , Q_{joint} , and Q_{cpl} and report cross-target distances instead of assuming equality.

2.2 Quadratic-loss identity

Equip aligned prediction arrays with the mean rowwise Euclidean inner product. For one-hot labels Y and Brier loss, or real-valued labels and squared loss,

$$\mathbb{E}_Z \|Y - P_Z\|^2 = \|Y - Q_\mu\|^2 + \mathbb{E}_Z \|P_Z - Q_\mu\|^2. \quad (2)$$

The cross term vanishes because $\mathbb{E}(P_Z - Q_\mu) = 0$. The second term is both quotient variance and the exact expected member-loss overhead under the quadratic score. This is a specialization of proper-score variance decompositions (Gruber & Buettner, 2023). It need not be large relative to total predictive loss and implies no accuracy or AUROC improvement.

2.3 Product fANOVA

Under a product measure, write

$$P_Z - Q_\mu = \sum_{\emptyset \neq u \subseteq [d]} f_u(z_u), \quad \mathbb{E}\langle f_u, f_v \rangle = 0 \quad (u \neq v). \quad (3)$$

Let $V_u = \mathbb{E}\|f_u\|^2$. The total quotient variance is exactly $\sum_u V_u$. This vector-valued product decomposition retains schema $\times$ RNG and higher-order interactions, which marginal perturbation cards discard. Functional ANOVA itself is established (Lengrich et al., 2020); our use is exact accounting in aligned prediction space.

3 OrbitCover

3.1 Interaction-balanced estimation

A budget- B design $\mathcal{D} = (Z_1, \dots, Z_B)$ estimates (1) by

$$\hat{Q}_{\mathcal{D}} = \frac{1}{B} \sum_{b=1}^B P_{Z_b}.$$

OrbitCover randomizes the level labels of a mixed-level OA over the declared factors. Every row is then marginally correct, so $\mathbb{E}_{\mathcal{D}} \hat{Q}_{\mathcal{D}} = Q_\mu$ for its finite target. A strength- t array contains every level combination equally often in every projection of at most t factors and cancels all f_u with $|u| \leq t$.

For exact accounting, let $W_{\mathcal{D}}$ be the random weight vector on the finite product, w its uniform vector, $C = \mathbb{E}[(W_{\mathcal{D}} - w)(W_{\mathcal{D}} - w)^\top]$, and Π_u the projector onto contrast subspace u . Independent randomization of factor labels makes C scalar on each product contrast subspace, giving

$$\mathbb{E}_{\mathcal{D}} \|\hat{Q}_{\mathcal{D}} - Q_\mu\|^2 = \sum_{u \neq \emptyset} \lambda_u V_u, \quad \lambda_u = \frac{|\Omega| \operatorname{tr}(C \Pi_u)}{\operatorname{rank}(\Pi_u)}. \quad (4)$$

IID draws have $\lambda_u = 1/B$; a strength- t array has $\lambda_u = 0$ for $|u| \leq t$. Equation (4) is an exact finite, mixed-level, vector-valued specialization of randomized-OA integration, not a new OA theorem.

3.2 Coupling is part of the method

OC2-independent balances the schema projection but assigns every fit a fresh unrelated master seed. OC2-coupled includes schema, initialization, and data-order factors in the strength-2 construction. Only the latter cancels their pairwise interaction components. The variants can also target different randomized procedures; coupling is therefore part of the estimand and the estimator, not bookkeeping.

Equations (2)–(4) guarantee unbiasedness for the declared finite target and describe estimator variance. They do not guarantee that a realized cover is invariant, that its quotient is more accurate than a canonical predictor, or that strength 2 beats simple random sampling in every prediction field.

4 Experimental design

Frozen endpoint and inference. All primary analyses were fixed before the final-closure outcomes. The protocol, configuration hashes, deviations, manifests, and regeneration commands accompany the submission. Classification uses aligned Brier residual; regression

Table 1: The frozen $B = 16$ showdown. The residual column is method-relative; the comparison is computational, not a same-estimand risk ordering.

Method	Mean residual	Ratio to canonical	Wins
OC2-coupled	0.000693	0.490	144/144
Canonical-independent	0.001415	1.000	—
OC1-independent	0.001512	1.068	7/144
IID-joint	0.001513	1.069	2/144
OC2-independent	0.001514	1.070	5/144
SRS-joint	0.001514	1.070	7/144

uses standardized prediction MSE. Estimator residual $\mathbb{E}\|\hat{Q} - Q\|^2$ is primary. Cached re-sampling draws estimate conditional expected residuals only. Datasets—not overlapping draws, rows, representatives, or fits—are the inferential unit. Headline intervals use 10,000 dataset-clustered bootstrap replicates.

Datasets and models. The primary panel has six classification datasets (Australian Credit, Bank Marketing, Credit Card Default, German Credit, HELOC, and Lending-Club) and six regression datasets (FREMtpl Claim Count, KDD17 Stock Return, Abalone, Kin8nm, Pol, and Puma32H), each under three deterministic splits. Experiment A caps train/validation/test at 2,048/512/512. We evaluate MLP, ResNet, FT-Transformer, and TabM, giving 144 primary cells. CatBoost and XGBoost are secondary first-split checks and do not enter the neural aggregate.

Actions and independent randomness. Schema actions combine four feature-block orders, up to four valid within-field category maps, and two target-ID maps for binary classification (one for regression). A unique signed 63-bit master seed is domain-separated by SHA-256 into initialization, dataloader, dropout, worker, preprocessing, and model-operation sub-seeds. No estimator or cell repeats an independent master seed.

The canonical reference averages 128 independent fits. The independent-joint reference averages eight independent predictions per schema action, giving 512 regression or 1,024 classification fits. The coupled reference exhausts the declared finite schema $\times$ initialization $\times$ order product. At $B \in \{4, 8, 16, 32, 64\}$ we compare canonical independent ensembling, IID joint sampling, schema sampling without replacement (SRS), strength-1 and strength-2 independent covers, and the coupled strength-2 cover.

Mechanism and convergence. A same-target 48-cell ablation crosses four datasets, four architectures, and three splits. It balances none, each factor, every factor pair, or all schema $\times$ initialization $\times$ order factors, always measuring residual to the same exact finite quotient. A separate six-dataset panel varies nested training size from 2,048 to the largest feasible sample and optimization from 20 to 200 epochs plus validation early-stopped convergence. Exact nuisance products are used at mandatory corners. A matched-function control transforms parameters so equivalent models agree initially to at most 10^{-6} .

5 Results

5.1 Coupled OrbitCover is efficient at small budget

At $B = 16$, OC2-coupled has lower residual than canonical-independent in 144/144 cells and all 12 dataset means (Figure 1). The equal-dataset reduction is 55.9%, the median cell reduction is 72.4%, and the dataset-clustered 95% interval is [38.7%, 73.8%]. Architecture reductions are 93.2% for MLP, 58.1% for ResNet, 39.1% for FT-Transformer, and 90.2% for TabM.

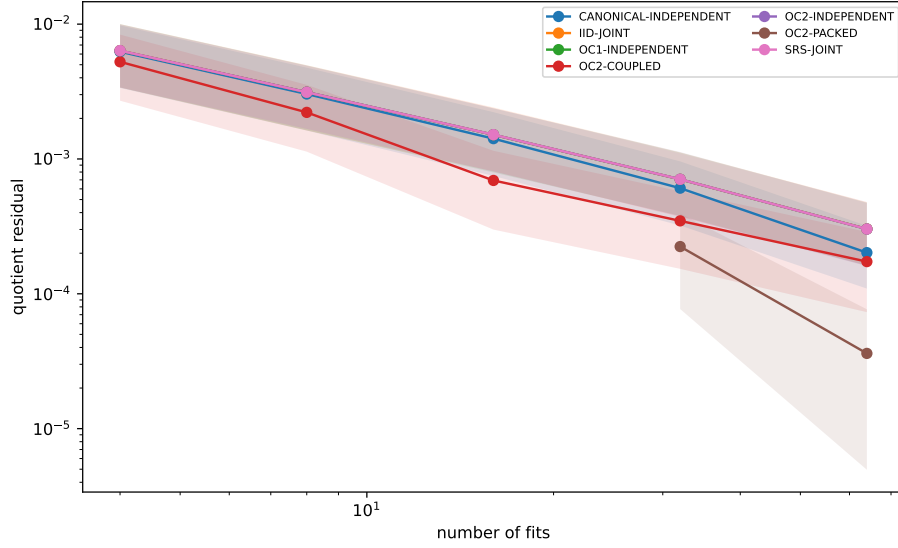


Figure 1: Frozen $B = 16$ independent-seed showdown across 144 neural cells. Residual is measured to each method’s declared reference; Section 5.2 audits the target difference.

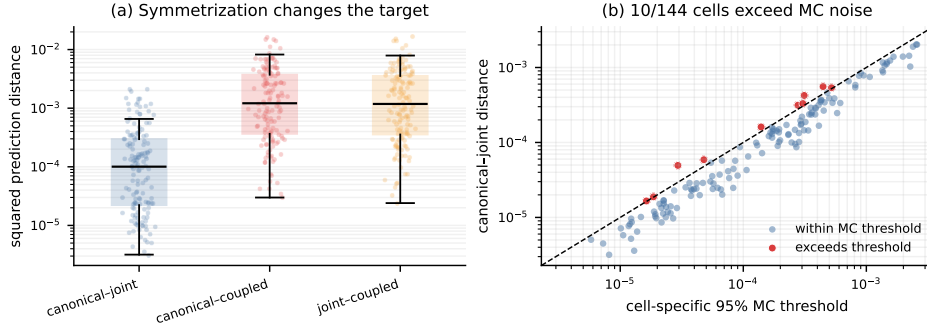


Figure 2: Target shift. Left: cellwise target distances. Right: the canonical-to-joint distance against its cell-specific Monte Carlo threshold.

5.2 Symmetrization changes the target

The mean squared distance between canonical and schema $\times$ independent expectations is 2.632×10^{-4} (median 1.005×10^{-4}), exceeding the cell-specific 95% Monte Carlo threshold in 10/144 cells. The finite coupled target is farther from the canonical target on average (2.494×10^{-3}). OrbitCover changes both what is averaged and how efficiently that average is estimated. We therefore never infer same-target dominance from Table 1.

5.3 Joint balance, rather than schema balance alone, drives the gain

OC2-independent beats canonical-independent in only 5/144 cells and no dataset mean. Its equal-dataset reduction is -7.0% , with a 95% interval of $[-7.8\%, -6.3\%]$. The same-target ablation gives the mechanism: all-factor strength-2 balance has a mean cellwise residual reduction of 47.6% and wins 48/48 cells. Schema alone reduces residual by 13.0%, initialization alone by 7.0%, and schema $\times$ initialization balance by 34.0%; all factors have the lowest residual. Schema $\times$ initialization fANOVA mass (0.01390) exceeds schema-only mass (0.00481), while joint higher-order mass (0.01155) remains material.

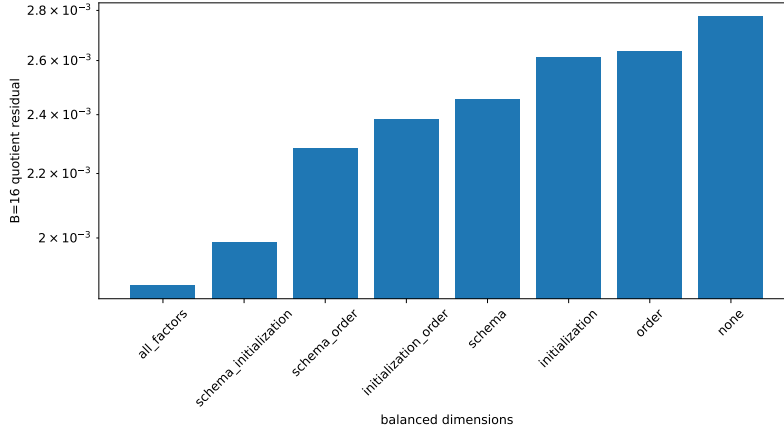


Figure 3: Same-target coupling ablation over 48 prospectively specified cells. Joint balance is not equivalent to schema balance plus arbitrary seeds.

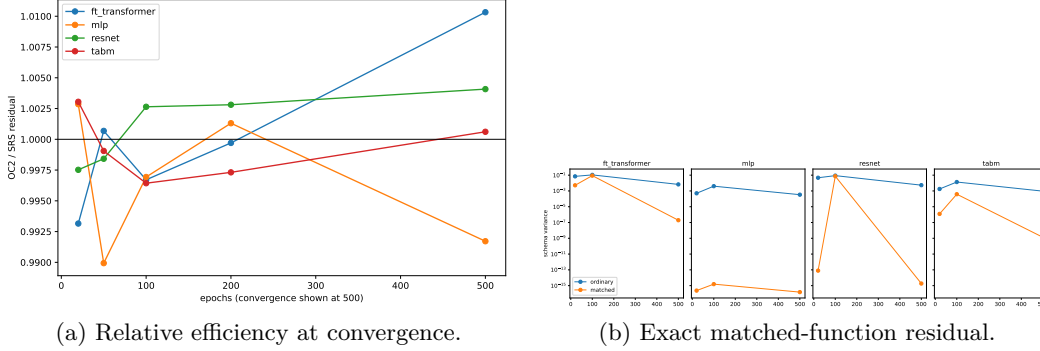


Figure 4: Training boundaries. The small-budget coupling effect is not a universal converged-path advantage.

5.4 The benefit is finite-budget and interaction-dependent

At convergence, the mean OC2/SRS residual ratio is 1.002: the strength-2 advantage does not persist on average. Nuisance variance remains nonzero, but its useful coupling changes with architecture and training. Main+pair fraction is only weakly associated with gain (Spearman $\rho = 0.139$, clustered interval $[-0.017, 0.320]$); higher-order fraction has $\rho = 0.258$ $[0.038, 0.415]$. Strength 3 recovers none of seven designated strength-2/SRS losses. Interaction order is a boundary condition, not a reliable per-cell oracle.

5.5 Exact matching narrows the optimizer-path claim

Under exact transformed initialization, residual is effectively zero for MLP (8.97×10^{-16}) and ResNet (8.41×10^{-14}), but initially remains 0.00262 for FT-Transformer and 4.19×10^{-6} for TabM. At convergence, the matched residual falls to 1.96×10^{-7} and 1.35×10^{-9} , respectively. Schema does not universally induce irreducible optimizer-path divergence. Architecture-specific tokenization, member structure, dropout, and minibatch dynamics remain the supported scope.

5.6 Validation fidelity does not imply held-out selection gain

In the prior exact selection panel, strength-2 winner agreement is 99.41% versus 96.69% for IID, and rank correlation is 98.64% versus 96.77%. Yet exact validation and test winners

agree in only 19/36 partitions. Held-out selected-test regret changes only from 0.005029 to 0.004906. Partition shift dominates the remaining selection error; we claim no practically meaningful held-out performance improvement.

6 Related work

Tabular models and evaluation. TabM (Gorishniy et al., 2025) shows that efficient ensembles are strong tabular baselines; we study which nuisance law such an ensemble integrates. TabPFN-v2, TabDPT, and TabSTAR learn broad tabular priors or representations (Hollmann et al., 2025; Ma et al., 2025; Arazi et al., 2025). TabArena and TabReD stress realistic benchmarking (Erickson et al., 2025; Rubachev et al., 2025). OrbitCover addresses variation inside a fixed data partition and does not solve temporal or population shift.

Symmetrization and multiplicity. Metamorphic testing already treats attribute and class-label permutations as valid supervised transformations (Xie et al., 2011). Frame averaging and learned probabilistic symmetrization reduce or learn group averages (Puny et al., 2022; Kim et al., 2023). OrbitCover instead samples complete retraining pipelines, returns an unbiased estimate rather than an exactly invariant realized network, and includes stochastic training factors. Neural training multiplicity itself is established (Summers & Dinneen, 2021); our focus is the product estimand.

Design and stable risk estimation. Randomized OAs and OA-based Latin hypercubes remove low-order ANOVA terms (Owen, 1992; Ai et al., 2016). OAs have also constructed low-correlation training ensembles (Johnson & Broatch, 2021), and orthogonal Monte Carlo couples continuous samples (Choromanski et al., 2019). Antithetic cross-validation couples response perturbations and gives strong contemporary risk theory (Liu et al., 2026; Chattopadhyay et al., 2026). These works rule out generic OA or antithetic novelty. OrbitCover acts on a finite semantic-pipeline nuisance product without perturbing responses or changing train/test membership.

7 Limitations and conclusion

The quotient is only as meaningful as its declared nuisance law. A uniform finite menu is transparent, but deployment may require a nonuniform or robust target. The 128-seed canonical and 512/1,024-fit joint references retain Monte Carlo error. Dataset-clustered intervals describe this 12-source panel rather than a distribution-free population. Evidence is tabular and limited to four primary neural architectures.

Quadratic prediction residual is the endpoint justified by (2). Accuracy, AUROC, calibration, fairness, and population-shift robustness can behave differently. The method is not universal: OC2-independent fails, average OC2/SRS efficiency vanishes at convergence, strength 3 does not repair the designated failures, and exact function matching nearly closes MLP and ResNet.

Within that boundary, the positive result is broad and mechanistically coherent. OrbitCover makes exact representation equivalences and training randomness into a finite semantic quotient, exposes its prediction-space interactions, and uses a coupled strength-2 design to estimate it efficiently at small budget. The practical lesson is conditional: design randomization around the interactions of the declared pipeline, and audit both target shift and finite-budget benefit.

AI use statement

Generative AI tools (OpenAI Codex) were used for literature discovery, hypothesis generation, confirmatory-protocol and experimental-code drafting, statistical-analysis assistance, figure-script preparation, manuscript drafting, and language editing. The authors selected and froze the confirmatory protocols before their outcomes, executed the experiments, inspected the generated artifacts, and checked the reported calculations and citations against

primary sources. All AI-assisted code is covered by the reported tests and artifact audit. The authors take responsibility for the final text, claims, code, and results.

Ethics statement

This work uses established public tabular benchmarks and introduces no human- subject data collection. Its main risk is evaluation overclaiming: an unstated nuisance law could obscure model multiplicity, while inappropriate symmetrization could waste compute or change a prediction target. We mitigate this by releasing action menus, semantic-alignment checks, cross-target distances, uncertainty, and all designated failures.

Reproducibility statement

The supplement contains the frozen protocol and hashes, all recorded deviations, model and dataset details, exact theorem statements, complete failure inventories, and regeneration commands. Machine-readable cell tables, summary JSON, vector PDF figures, and the persistent fit registry accompany the source. A read-only audit verifies 140,592 unique fit keys, 1,128 aligned prediction arrays, 125,616 independent seed records, all mandatory manifests, and 116/116 tests.

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A Proof details

Quadratic ambiguity identity. Write $Y - P_Z = (Y - Q_\mu) + (Q_\mu - P_Z)$, expand the squared Hilbert norm, and take expectations. The cross term is zero because $\mathbb{E}(P_Z - Q_\mu) = 0$, proving (2). The same argument with $\hat{Q}_{\mathcal{D}}$ shows that cover residual is exact expected Brier/MSE overhead whenever the design is unbiased for Q_μ .

Product fANOVA filter. Define f_u recursively by subtracting every lower-order conditional expectation from $\mathbb{E}[P_Z \mid Z_u]$. Under a product measure these components are mutually orthogonal. Randomized factor-label permutations make the design covariance commute with each factor’s symmetric-group action, so it is scalar on each tensor-product contrast subspace. Projection and trace normalization give (4). Strength t makes every empirical marginal through order t uniform, hence annihilates those subspaces.

B Additional experimental details

Architectures. All primary cells use the frozen MLP, ResNet, FT-Transformer, and TabM widths, AdamW, batch size 256, learning rate 10^{-3} , weight decay 10^{-4} , and 20 epochs unless the scale experiment explicitly changes the budget. The convergence arm uses at most 500 epochs, patience 30, relative minimum improvement 10^{-4} , and restores the best checkpoint.

Reference construction. Every fit is keyed by dataset, split, model/configuration hash, schema hash, master RNG seed or declared finite init/order pair, training size, budget, and matched arm. Prediction files are registered only after shape, alignment, and finiteness checks. Resampling uses 512 deterministic estimator draws per cell and method; overlaps are never counted as independent evidence.

Compute. The audited registry represents 140,592 fits and 232.005 summed local fit-hours (231.802 GPU and 0.203 CPU). These measurements document execution and reproducibility, not portable latency, energy, or monetary speedup.

C Complete headline ledger

Claim	Audited value
OC2-coupled vs canonical-independent	144/144 cell wins; 12/12 dataset wins; 55.9% [38.7%, 73.8%]
OC2-independent vs canonical-independent	5/144 cell wins; 0/12 dataset wins; -7.0% [-7.8%, -6.3%]
Canonical-to-independent-joint distance	Mean 2.632×10^{-4} ; 10/144 over cellwise MC threshold
Same-target all-factor ablation	48/48 wins; 47.6% mean cellwise reduction
Convergence boundary	Mean OC2/SRS residual ratio 1.001683
Interaction prediction	Main+pair $\rho = .139$ [-.017, .320]; higher $\rho = .258$ [.038, .415]
Selection boundary	Exact validation/test winner agreement 19/36; regret .005029 to .004906

D Excluded claims

The evidence does not support universal superiority over independent seed ensembling, a common expectation across methods, persistent convergence efficiency, universal irreducible optimizer-path dependence, strength-3 dominance, meaningful held-out accuracy improvement, or portable systems speedup. OAs, fANOVA filtering, group averaging, antithetic dependence, and proper-score variance identities are cited as prior art rather than claimed as inventions.